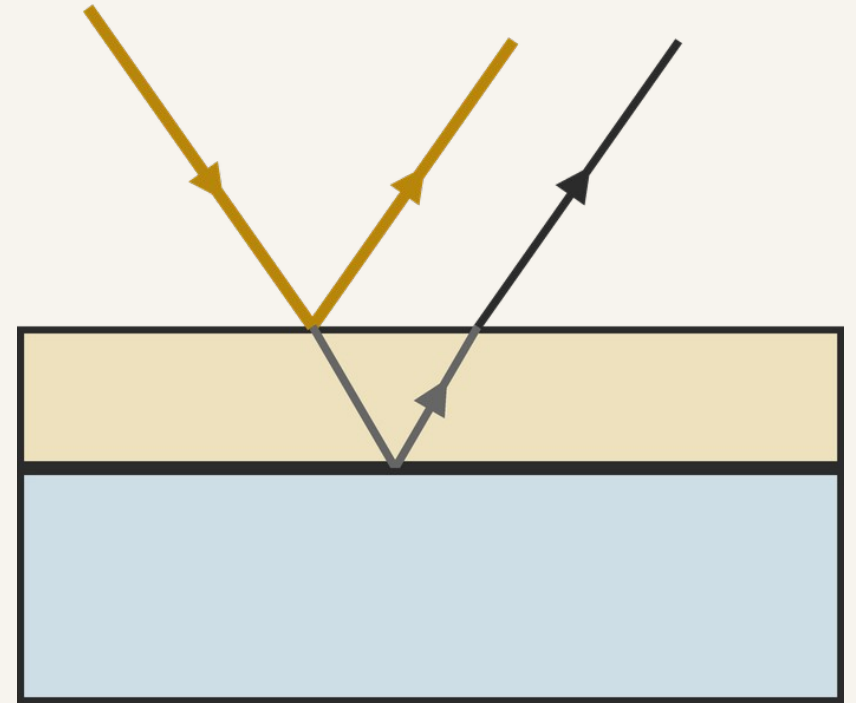


# Division of Amplitude

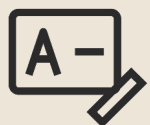
---

Interference part 2: thin films, coated lenses and air wedges.

Advanced Higher Physics



# What the pictures mean



**Show me boards**  
Hold up your board.



**Write it**  
Write it in your jotter.



**Turn and talk**  
Talk to your partner.



**Hands down**  
No hands up. I pick.



**Hands up**  
Hand up to answer.



**Silent task**  
Work alone, in silence.



**Jotter notes**  
Copy it into your jotter.



**Cut and stick**  
Cut out and stick in.

## LEARNING INTENTION

To explain how one wave split at two surfaces makes interference, in thin films, coated lenses and air wedges.

## By the end of this deck...

- 1 I can use  $\text{opd} = n \times \text{gpd}$  to solve problems.
- 2 I can state when reflection gives a phase change of  $\pi$ .
- 3 I can explain interference by division of amplitude.
- 4 I can use  $\text{opd} = m\lambda$  or  $(m + \frac{1}{2})\lambda$ , and  $\Delta x = \lambda l / 2d$ .
- 5 I can derive  $d = \lambda / 4n$  for a coated lens.

# Recap – from Division of Wavefront

## Coherent

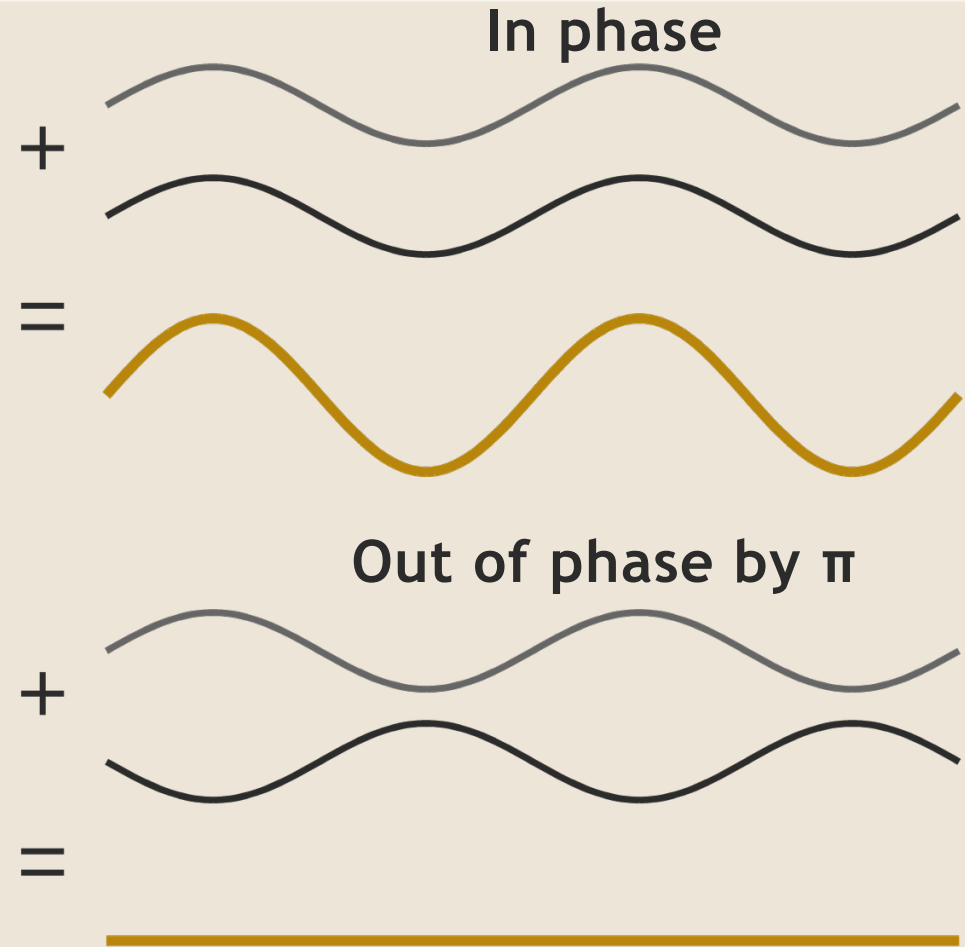
Constant phase relationship.

## Path difference

Max:  $m\lambda$ . Min:  $(m + \frac{1}{2})\lambda$ .

## Phase difference

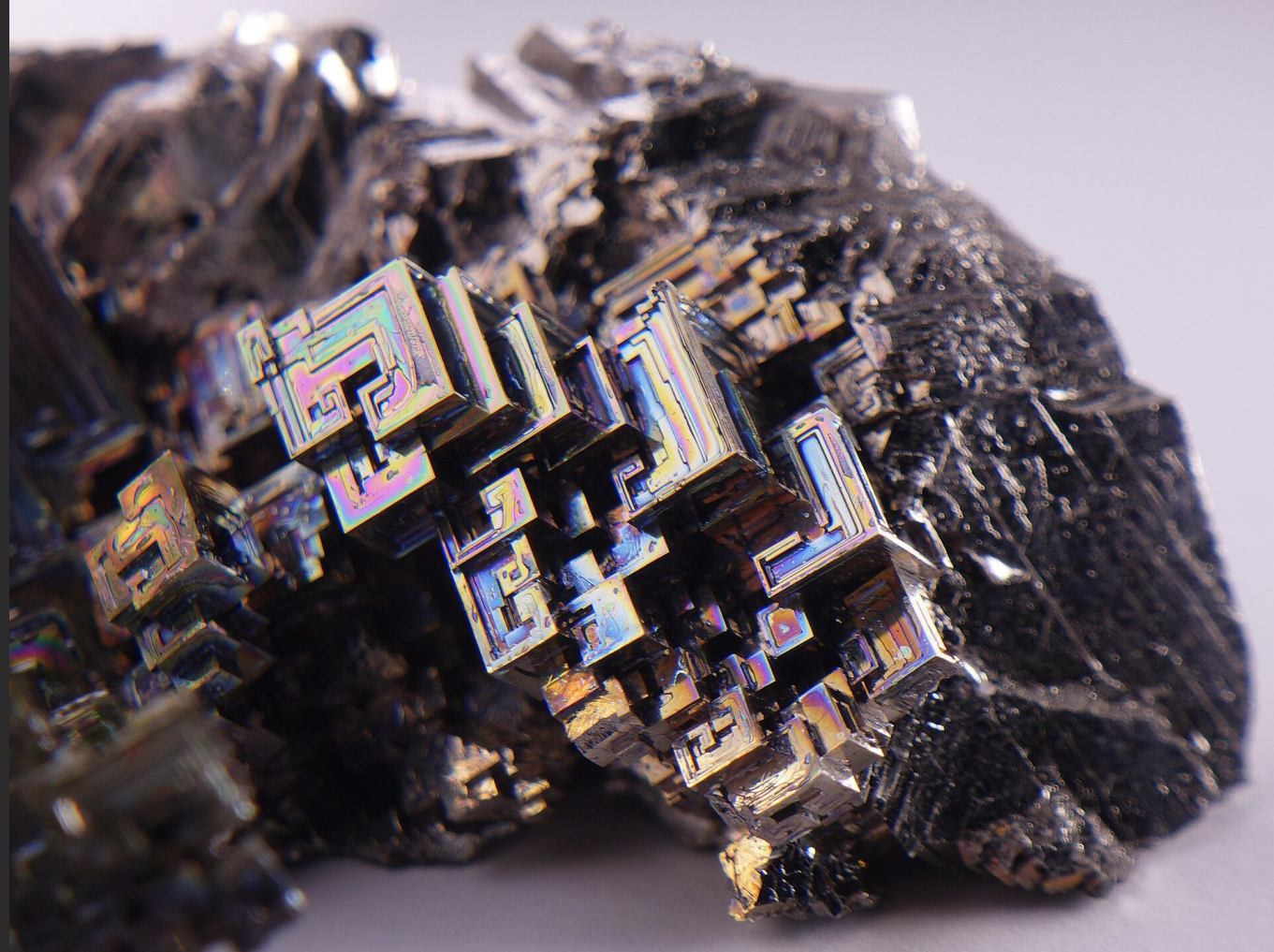
$\varphi = 2\pi x/\lambda$ . Half a  $\lambda$  is  $\pi$ .



# Daily Review



- 1 State what is meant by coherent waves.
- 2 State the path difference for destructive interference.
- 3 Write the relationship between phase difference and path difference.
- 4 Light goes from air into glass. What happens to its wavelength?
- 5 State what is meant by refractive index.



This metal crystal has no paint and no dye. Why does it shine with every colour?

## PART 1

# Why does light go further in glass?

It fits more waves into the same distance.

## By the end of this part...

- 1 I can use  $\text{opd} = n \times \text{gpd}$  to solve problems.
- 2 I can state the phase change on reflection from a denser medium.
- 3 I can state the phase change on reflection from a less dense medium.

# Optical path length

## Refractive index

$$n = c \div v.$$

## In glass

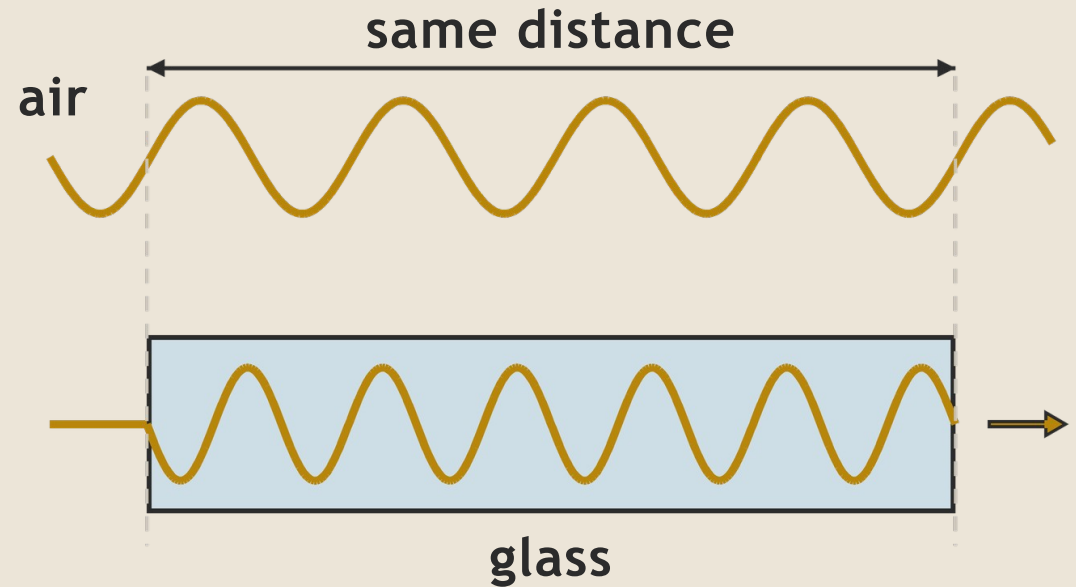
$\lambda$  shrinks to  $\lambda/n$ .

## Same distance

$n$  times as many waves.

## Optical path

$n \times$  geometrical path.



# Optical path difference

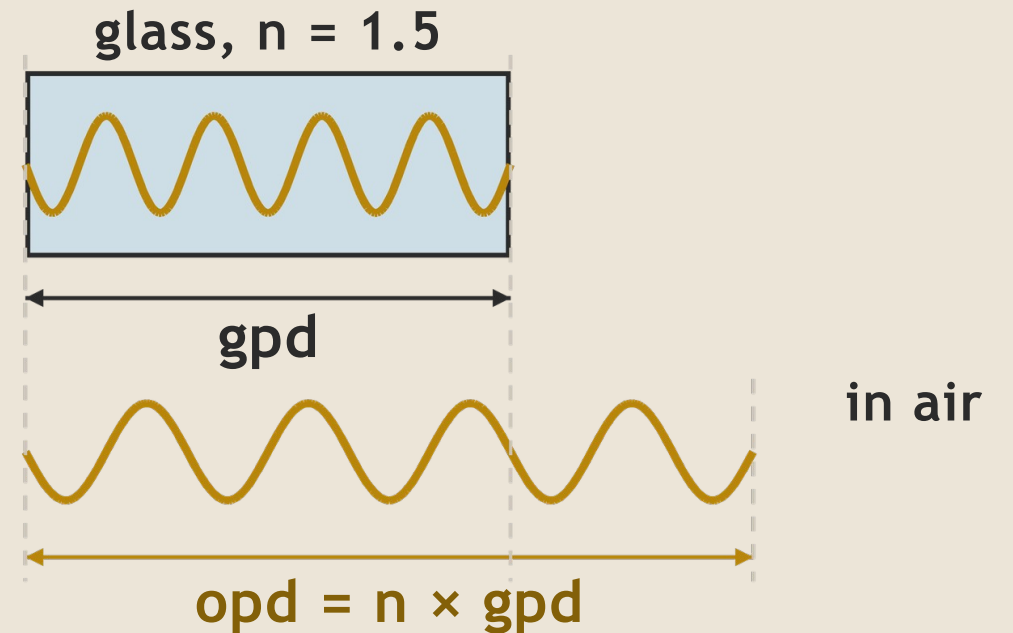


$$opd = n \times gpd$$

opd: optical path difference (m)

n: refractive index (no unit)

gpd: geometrical path difference (m)



# At a boundary – incident



## Incident ray

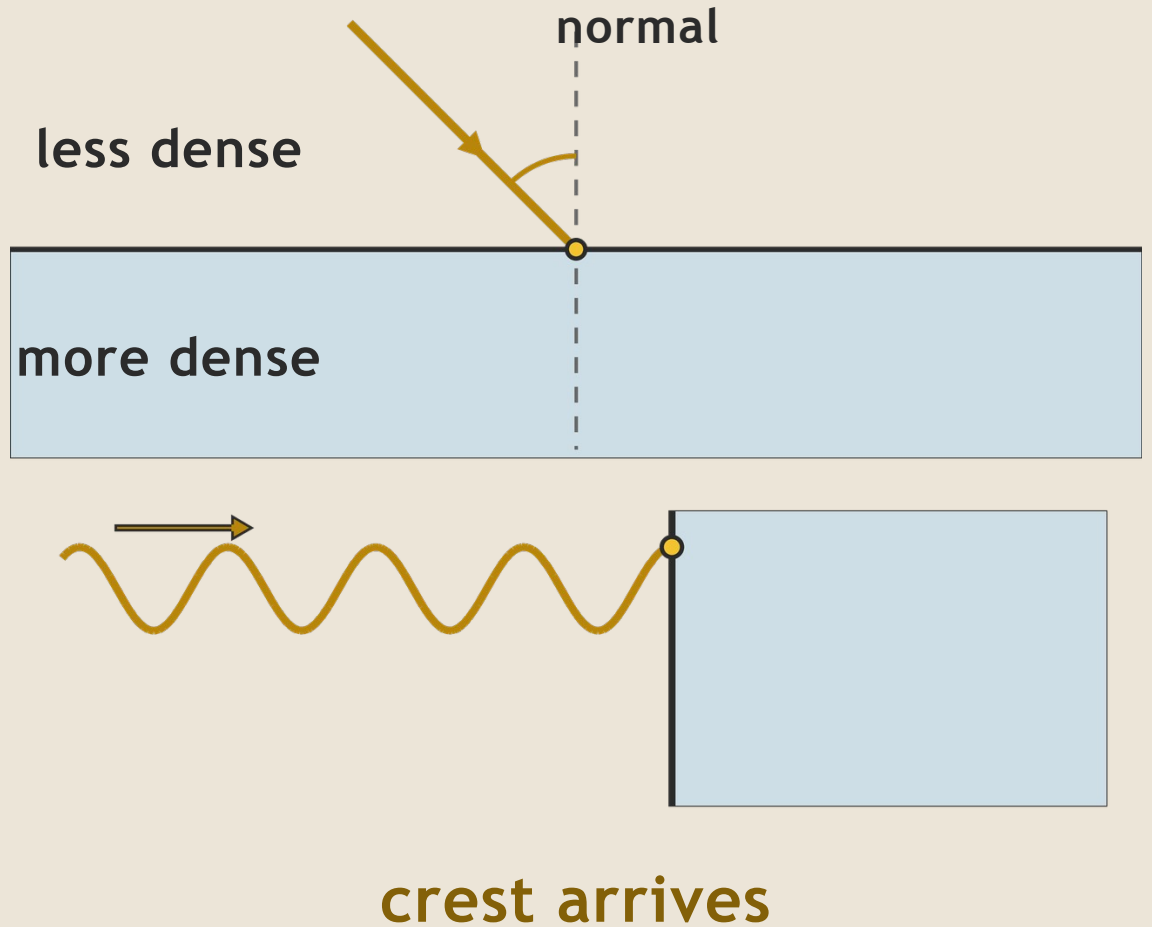
Arrives at the boundary.

## Normal

At  $90^\circ$  to the boundary.

## Phase

Reference: a crest arrives.



# At a boundary – reflected off more dense



## Reflected

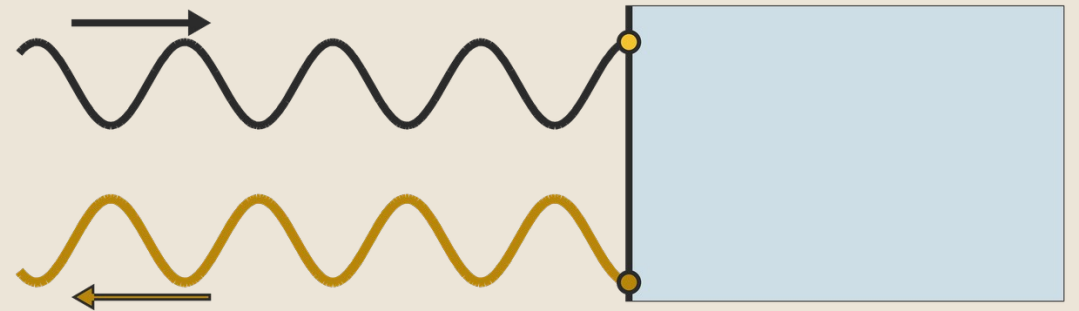
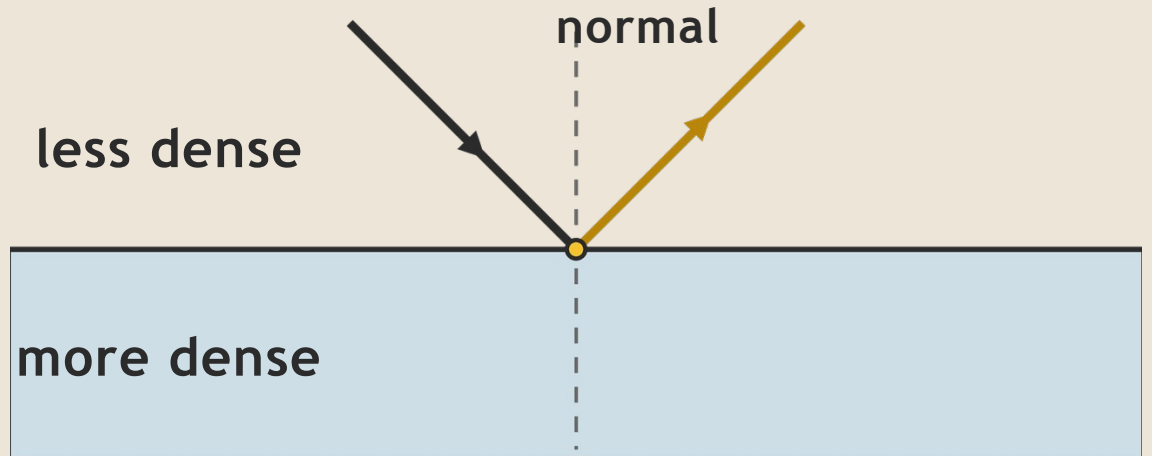
Bounces back off the boundary.

## Angles

Reflection = incidence.

## Phase change = $\pi$

A crest comes back as a trough.



phase change =  $\pi$

# At a boundary – reflected off less dense



## Reflected

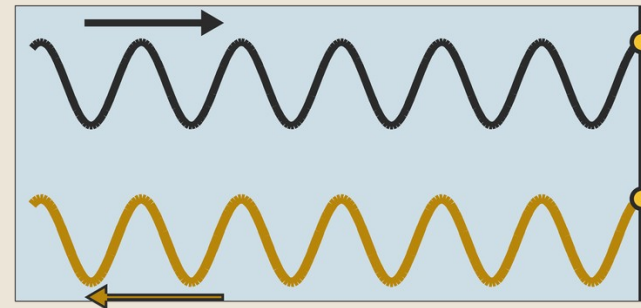
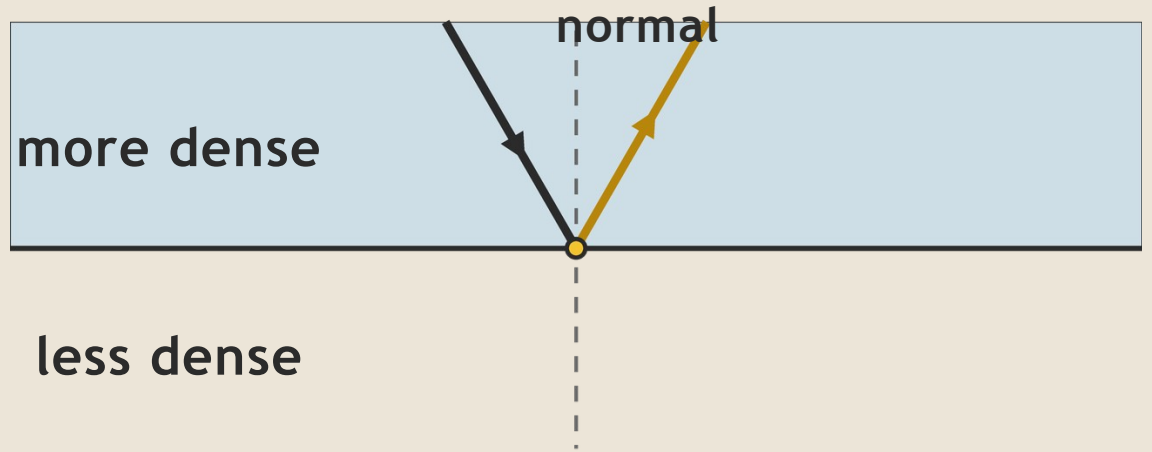
Bounces back off the boundary.

## Angles

Reflection = incidence.

## Phase change = 0

A crest comes back as a crest.



phase change = 0

# At a boundary – refracted into more dense



## Refracted

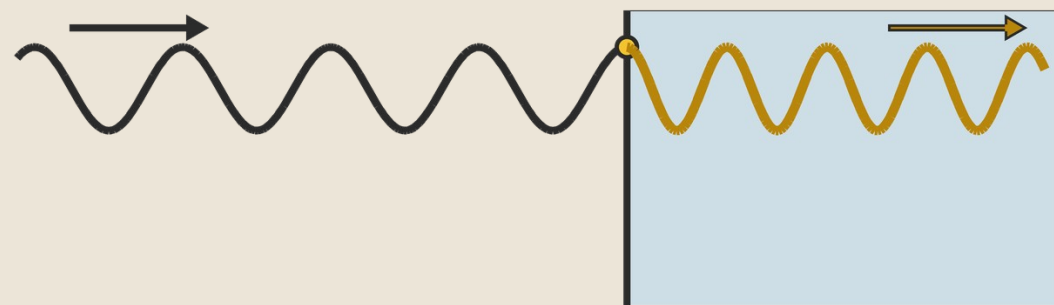
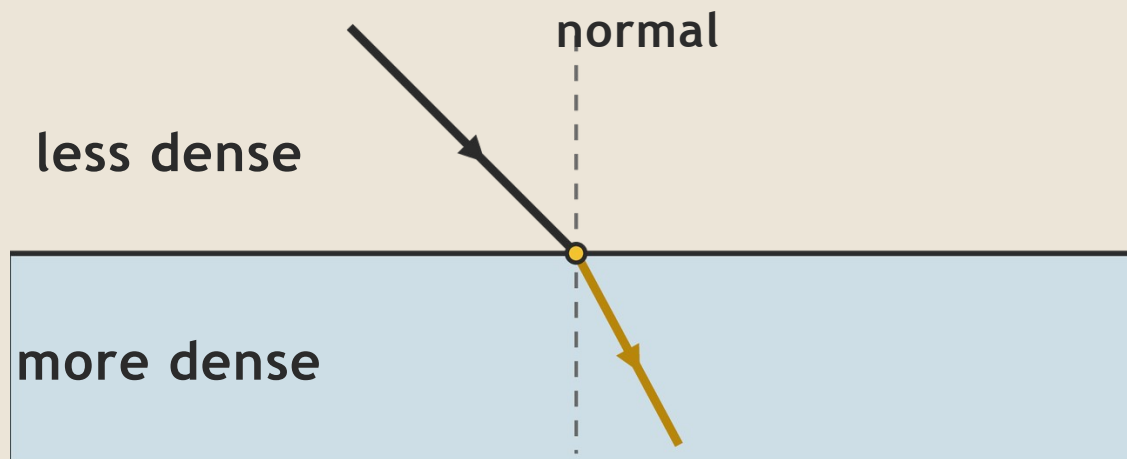
Slows. Bends towards normal.

## Wavelength

Shorter:  $\lambda/n$ . Same frequency.

## Phase change = 0

A crest goes on as a crest.



phase change = 0

# At a boundary – refracted into less dense



## Refracted

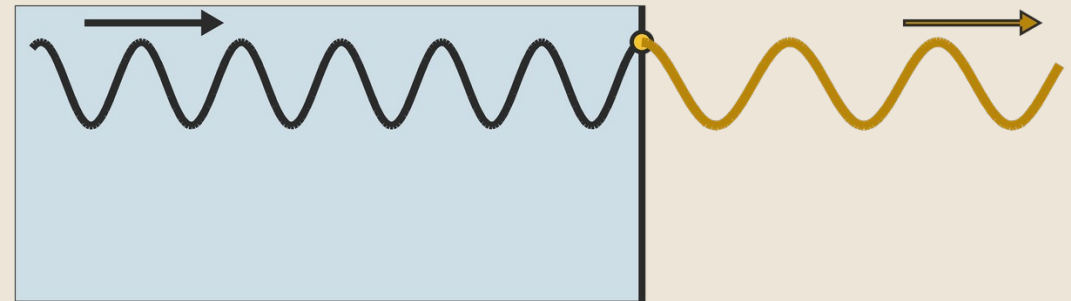
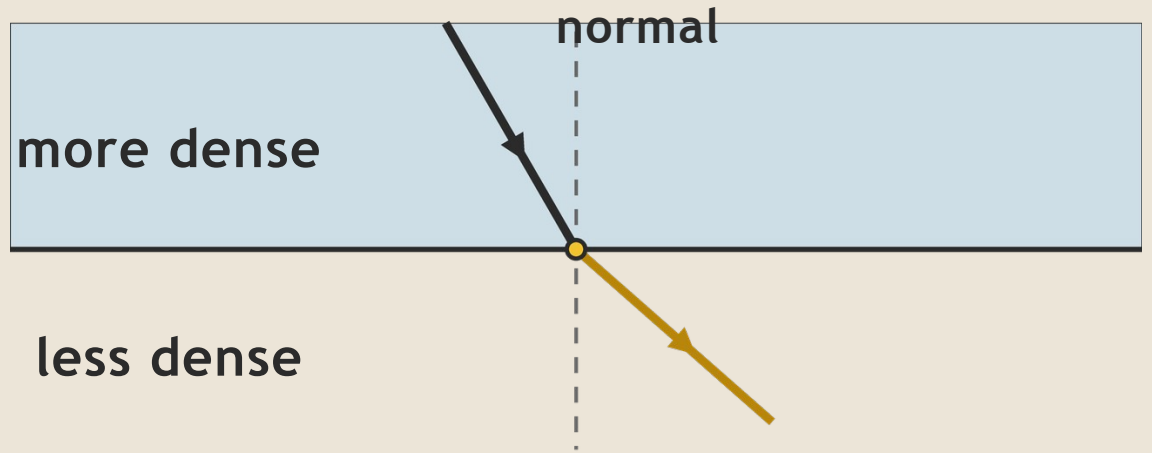
Speeds up. Bends away.

## Wavelength

Longer. Same frequency.

## Phase change = 0

A crest goes on as a crest.



phase change = 0

# Phase change – summary



**Reflected off more dense**

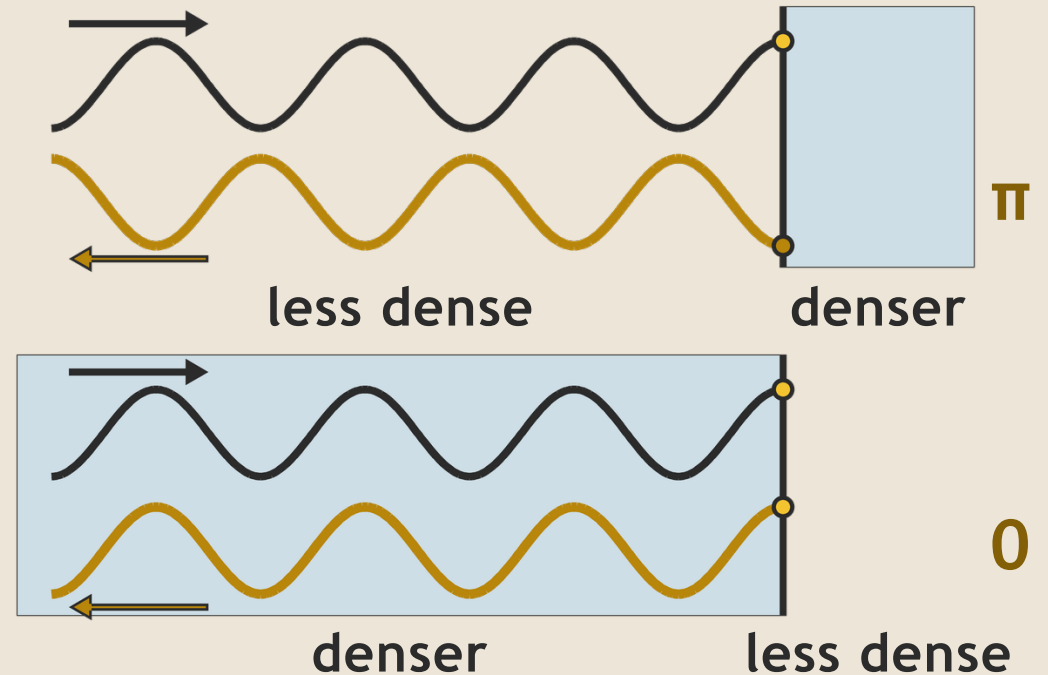
Phase change =  $\pi$ .

**Reflected off less dense**

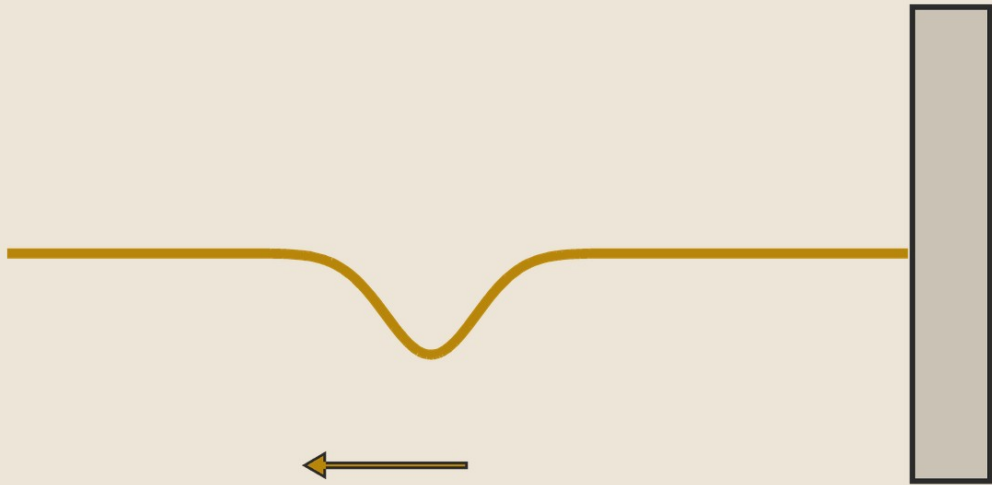
Phase change = 0.

**Refracted, either way**

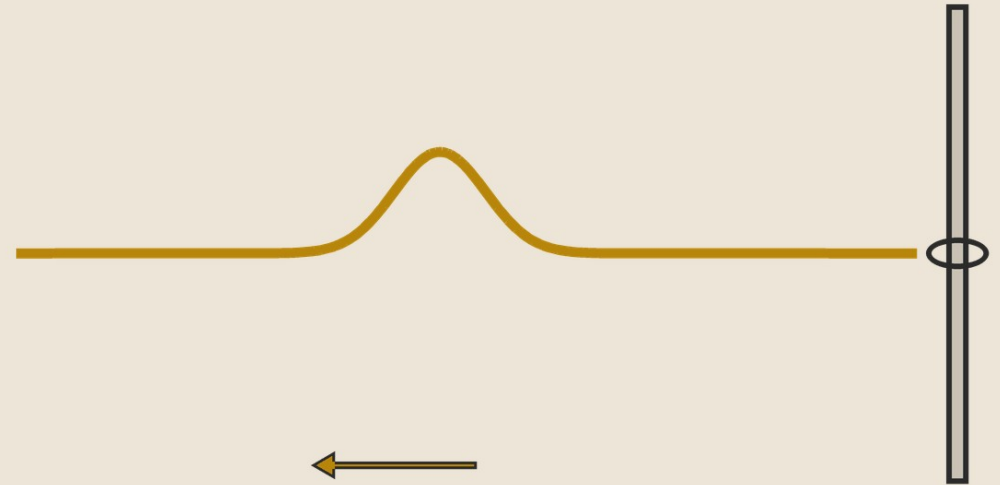
Phase change = 0.



## Phase change – other ways to see it



Fixed end: inverted



Free end: upright

## Example answer – coating less dense than glass

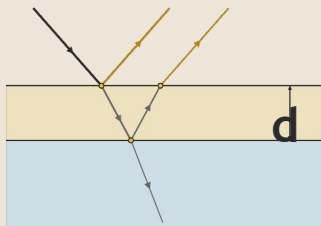
A glass lens ( $n = 1.52$ ) has a coating of  $n = 1.38$ .

State the phase change for light reflected at:

- (i) the air-coating boundary
- (ii) the coating-glass boundary. (2)

$n: 1.00 \rightarrow 1.38$

$n: 1.38 \rightarrow 1.52$



(i)  $\pi$  rad (1)

(ii)  $\pi$  rad (1)

Unit optional, but must be right.

## Example answer – coating denser than glass

A glass plate ( $n = 1.50$ ) has a coating of  $n = 2.40$ .

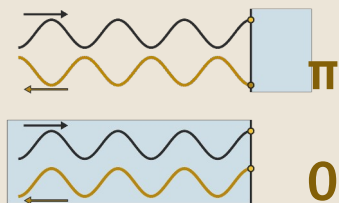
State the phase change for light reflected at:

(i) the air-coating boundary

(ii) the coating-glass boundary. (2)

$n: 1.00 \rightarrow 2.40$

$n: 2.40 \rightarrow 1.50$



(i)  $\pi$  rad (1)

(ii) no phase change (1)

Accept 0 rad for (ii).

## Look back — by the end of this part...

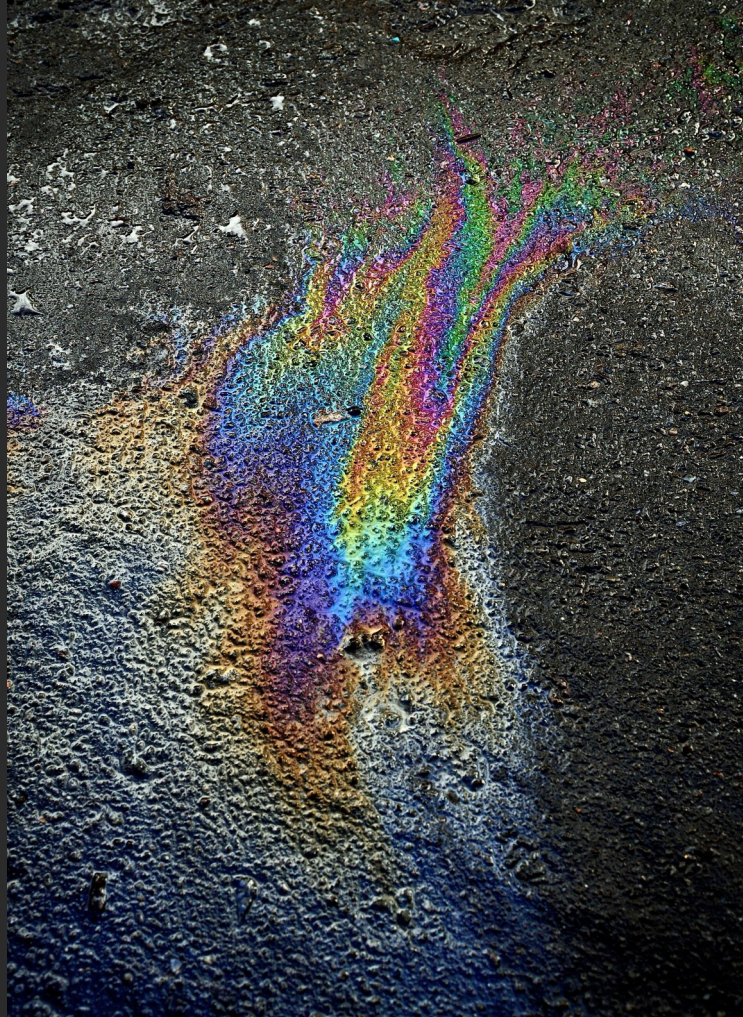


- 1 I can use  $\text{opd} = n \times \text{gpd}$  to solve problems.
- 2 I can state the phase change on reflection from a denser medium.
- 3 I can state the phase change on reflection from a less dense medium.

# Daily Review



- 1 State the relationship for optical path difference.
- 2 Light in air reflects off glass. State the phase change.
- 3 Light in glass reflects off air. State the phase change.
- 4 State what is meant by division of wavefront.
- 5 State the condition for constructive interference.



Oil on a wet road. Where do the colours come from?

## PART 2

# Why does a thin film shine with colour?

Two reflections of one wave interfere.

## By the end of this part...

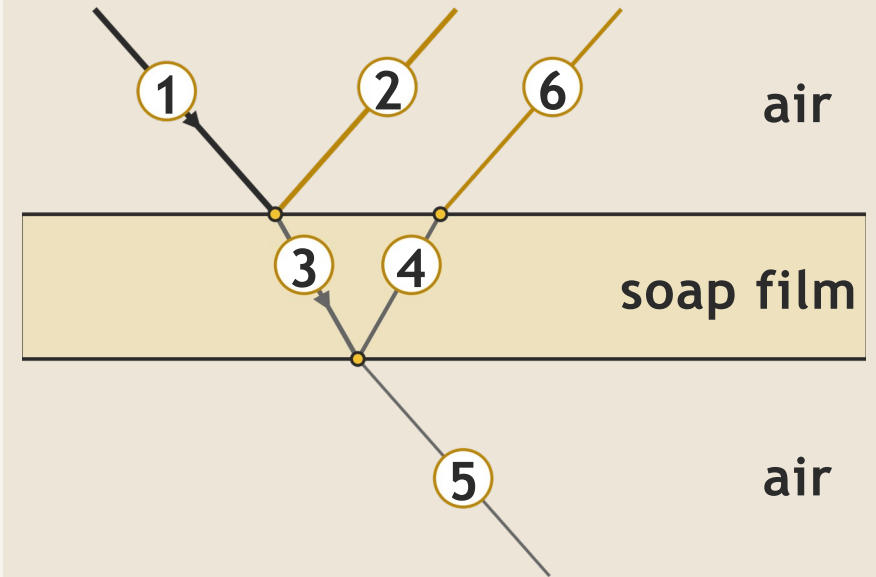
- 1 I can explain interference by division of amplitude.
- 2 I can use  $\text{opd} = m\lambda$  or  $(m + \frac{1}{2})\lambda$  to solve problems.
- 3 I can derive  $d = \lambda/4n$  for a coated lens.

# Division of amplitude



Rays 2 and 6 interfere: coherent.

| ray                         | phase change |
|-----------------------------|--------------|
| 1 incident                  | —            |
| 2 reflected off more dense  | $\pi$        |
| 3 refracted into more dense | 0            |
| 4 reflected off less dense  | 0            |
| 5 refracted into less dense | 0            |
| 6 refracted into less dense | 0            |



## Bright or dark?



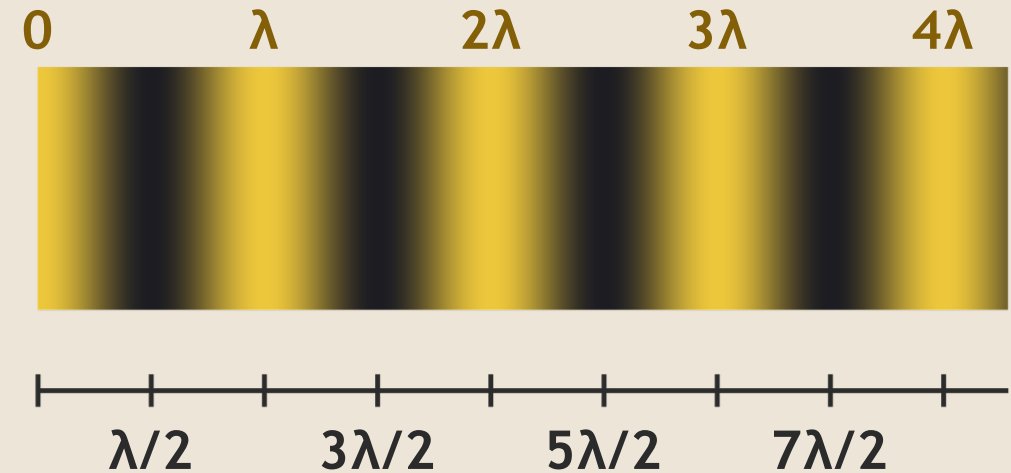
$$opd = m\lambda$$

$$opd = (m + \frac{1}{2})\lambda$$

$m = 0, 1, 2 \dots$

opd: optical path  
difference (m)

$\lambda$ : wavelength (m)



One  $\pi$  shift on reflection swaps  
bright and dark.

# A soap film



**gpd**

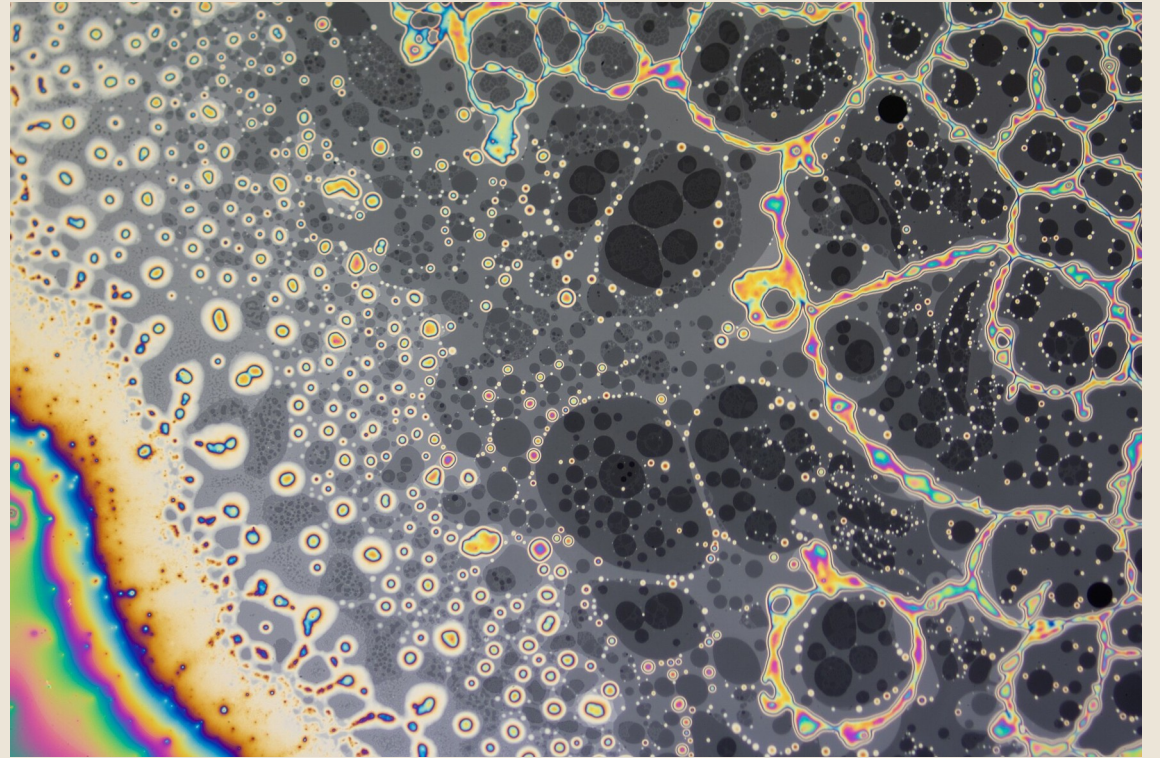
2d: down and back up.

**Top surface**

$\pi$ . Bottom surface: none.

**Thinnest film**

opd near 0, so dark.



Black: thinnest film.

# Why the colours change



## Film drains

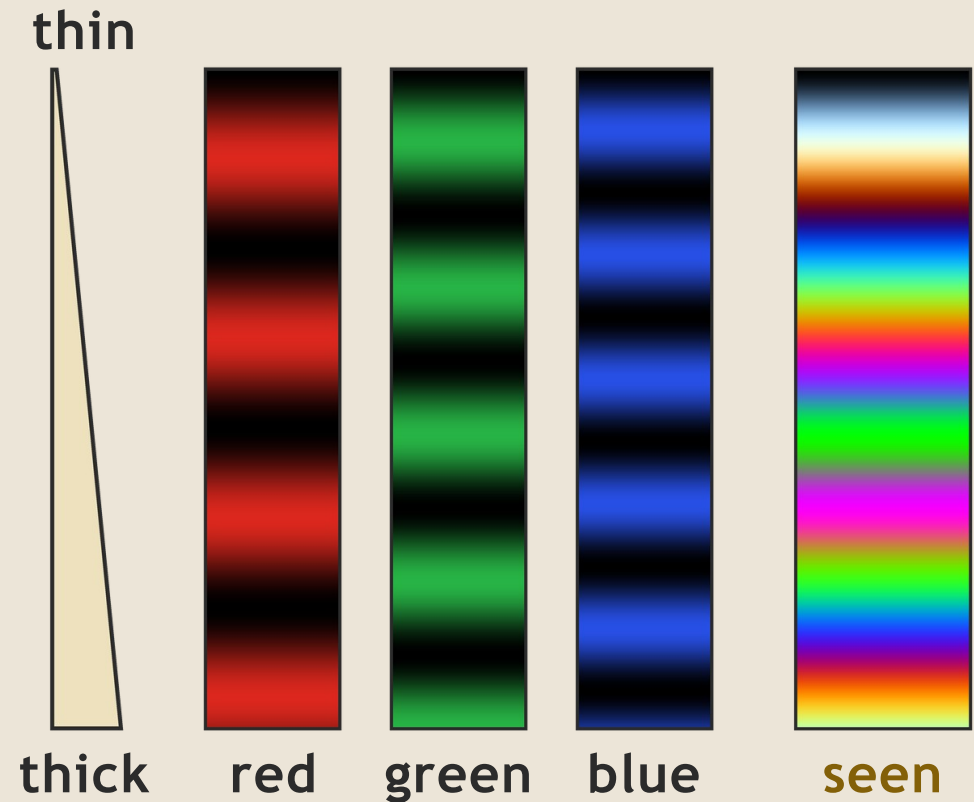
d grows down the film.

## Each colour

Bright at  $2nd = (m + \frac{1}{2})\lambda$ .

## Colour seen

Mix of the bright colours.



## Worked example 1 – optical path difference

Light of wavelength 589 nm reflects from a soap film ( $n = 1.33$ ). At P the film is  $2.80 \times 10^{-6}$  m thick. Calculate the opd between the two reflected rays. (3)

$$n = 1.33$$

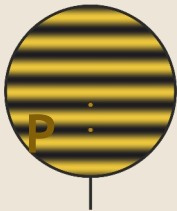
$$gpd = 2 \times 2.80 \mu\text{m}$$

$$\text{opd} = ?$$

$$\text{opd} = n \times gpd$$

$$\text{opd} = 1.33 \times (2 \times 2.80 \times 10^{-6})$$

$$\underline{\text{opd} = 7.45 \times 10^{-6} \text{ m}}$$



## Worked example 2 – the next dark band

P is dark. The next dark area is at Q, where the film is slightly thinner. Determine the opd at Q. (1)

$$P: 7.45 \times 10^{-6} \text{ m}$$

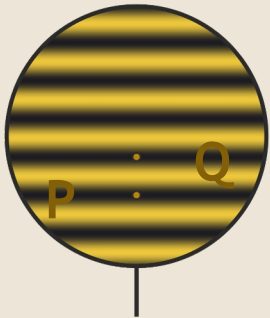
$$\lambda = 589 \times 10^{-9} \text{ m}$$

$$Q: \text{opd} = ?$$

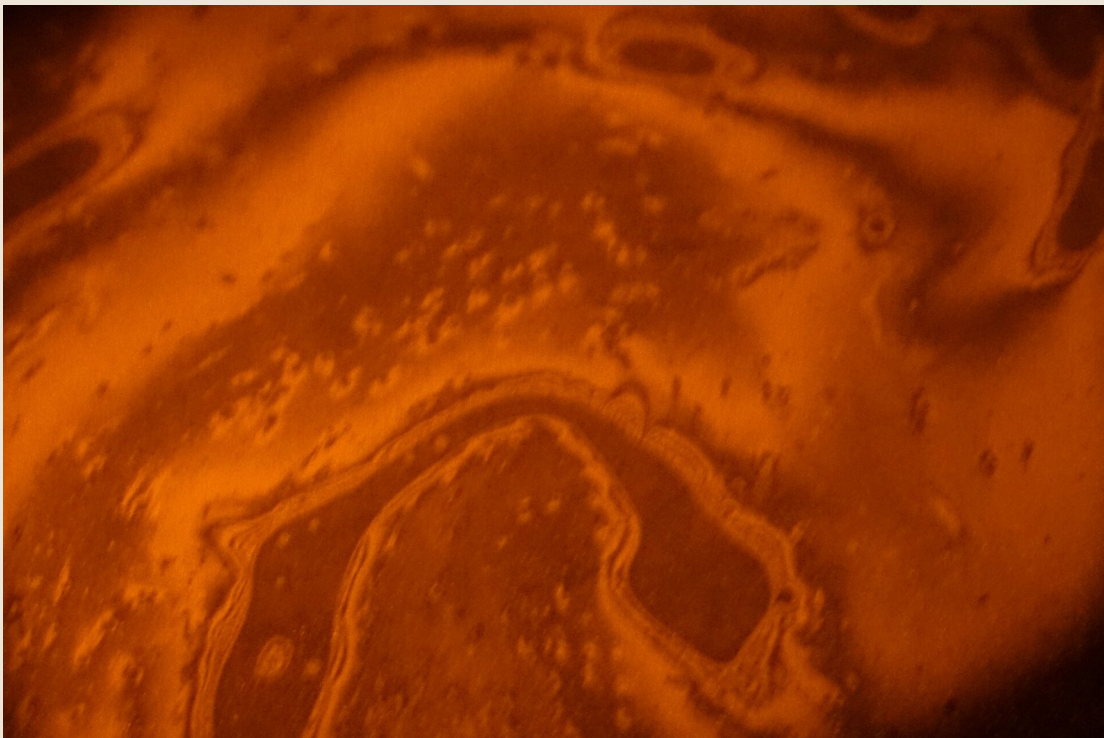
Next dark band: one  $\lambda$  less.

$$\text{opd} = 7.45 \times 10^{-6} - 589 \times 10^{-9}$$

$$\underline{\text{opd} = 6.86 \times 10^{-6} \text{ m}}$$



## One colour, then white light



One  $\lambda$ : bright and dark



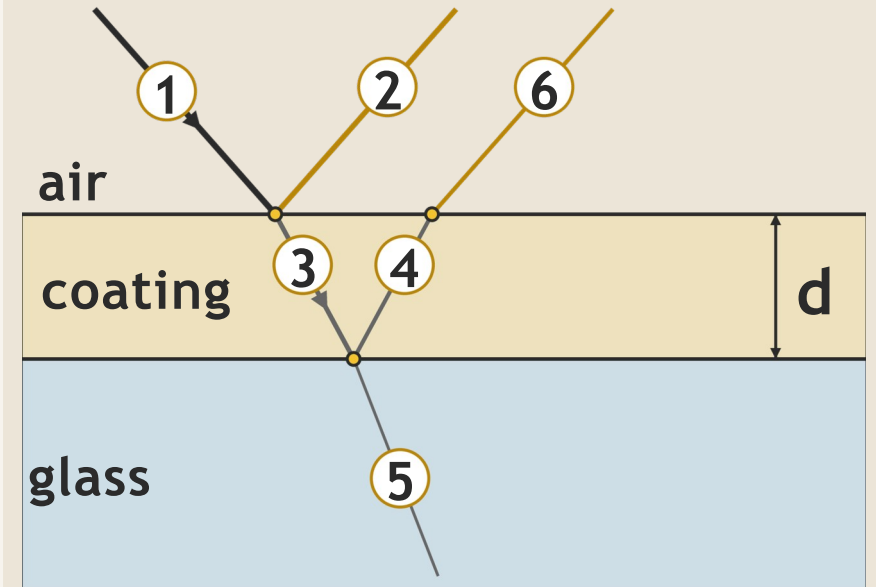
White light: colours

# Coated (bloomed) lenses



Right d: rays 2 and 6 cancel.

| ray                         | phase change |
|-----------------------------|--------------|
| 1 incident                  | —            |
| 2 reflected off more dense  | $\pi$        |
| 3 refracted into more dense | 0            |
| 4 reflected off more dense  | $\pi$        |
| 5 refracted into more dense | 0            |
| 6 refracted into less dense | 0            |



# Deriving $d = \lambda/4n$ (1 of 3)



**MANDATORY DERIVATION — you must reproduce every line**

$$n_{\text{lens}} > n > 1$$

Coating between air and lens.

$$\phi_X = \pi$$

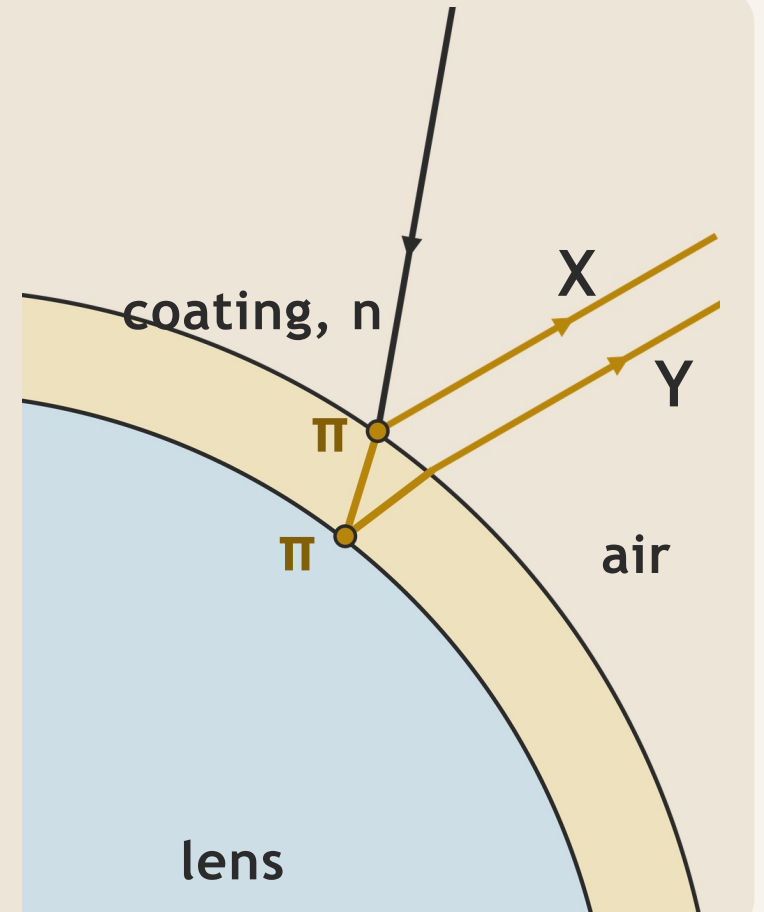
X: off the coating, more dense.

$$\phi_Y = \pi$$

Y: off the lens, more dense.

$$\phi_Y - \phi_X = 0$$

No net phase change.



## Deriving $d = \lambda/4n$ (2 of 3)



**MANDATORY DERIVATION — you must reproduce every line**

$$opd = n \times gpd$$

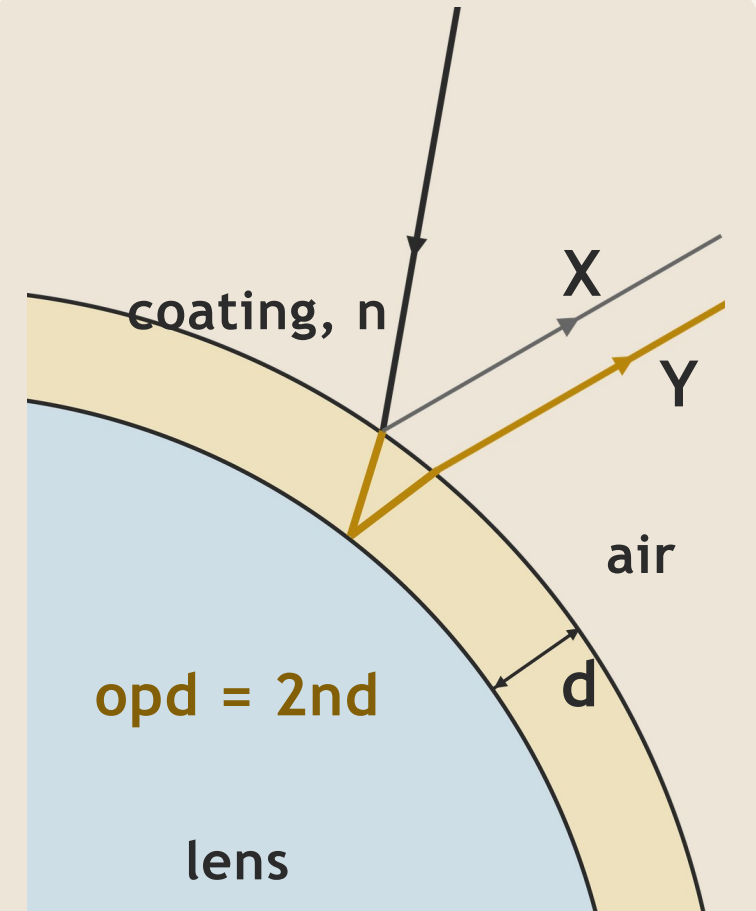
Near normal incidence.

$$gpd = 2d$$

Y crosses the coating twice.

$$opd = 2nd$$

Substitute  $gpd = 2d$ .



## Deriving $d = \lambda/4n$ (3 of 3)



**MANDATORY DERIVATION — you must reproduce every line**

$$2nd = (m + \frac{1}{2})\lambda$$

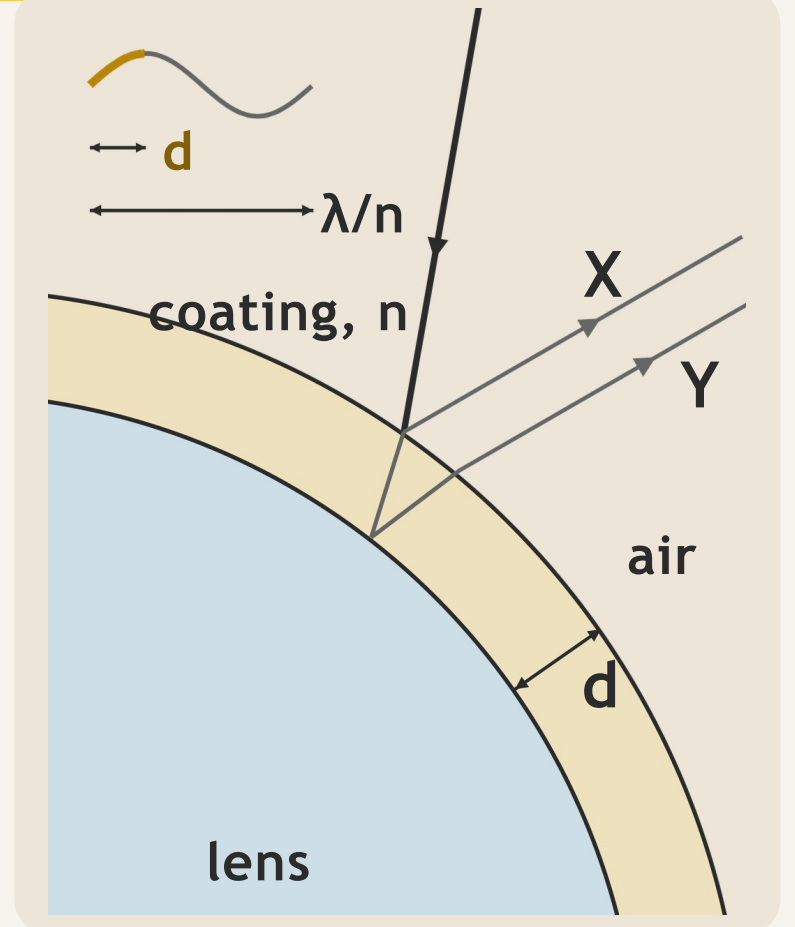
$$2nd = \lambda/2$$

$$d = \lambda/4n$$

X and Y cancel:  
destructive.

Thinnest:  $m = 0$ .

Divide by  $2n$ .



# Coating thickness



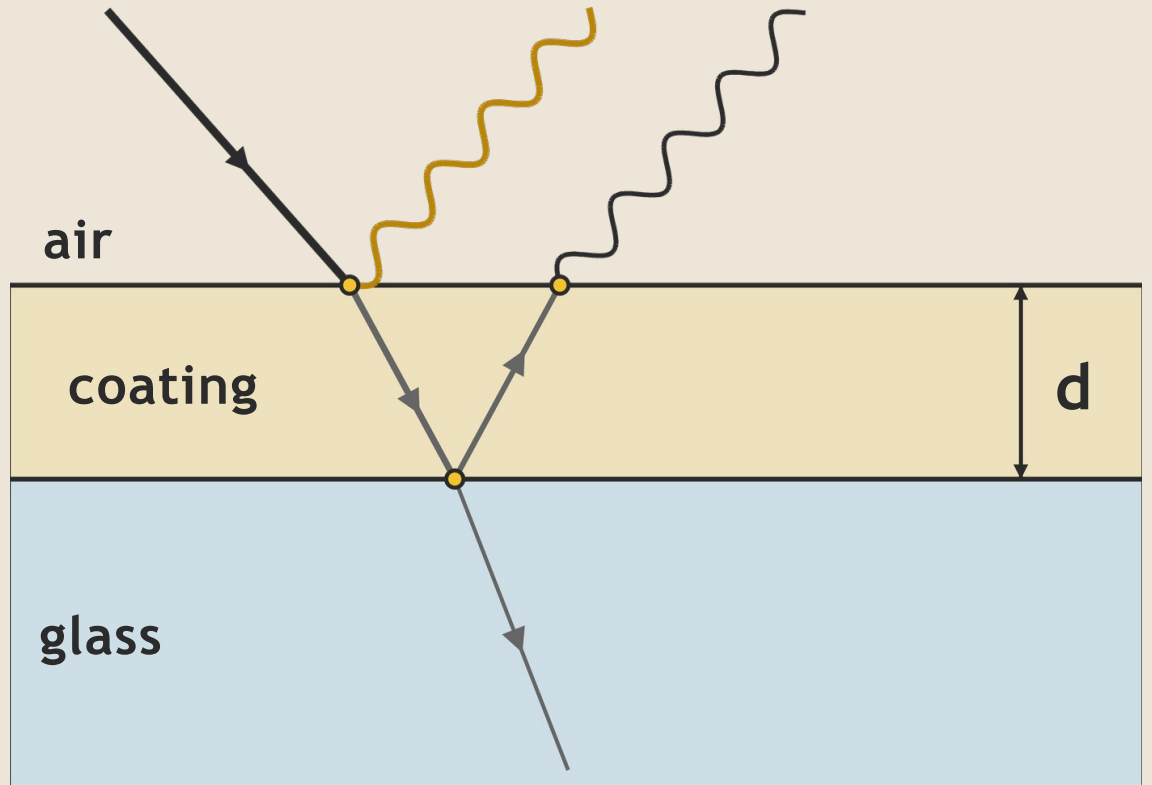
$$d = \lambda/4n$$

$d$ : coating thickness (m)

$\lambda$ : wavelength in air (m)

$n$ : refractive index of the coating

Works for one wavelength only.



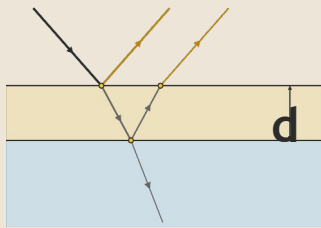
## Worked example 3 – coating thickness

A coating ( $n = 1.38$ ) makes a lens non-reflecting for light of wavelength 560 nm. Calculate the minimum thickness of the coating. (3)

$$\lambda = 560 \times 10^{-9} \text{ m}$$

$$n = 1.38$$

$$d = ?$$



$$d = \lambda/4n$$

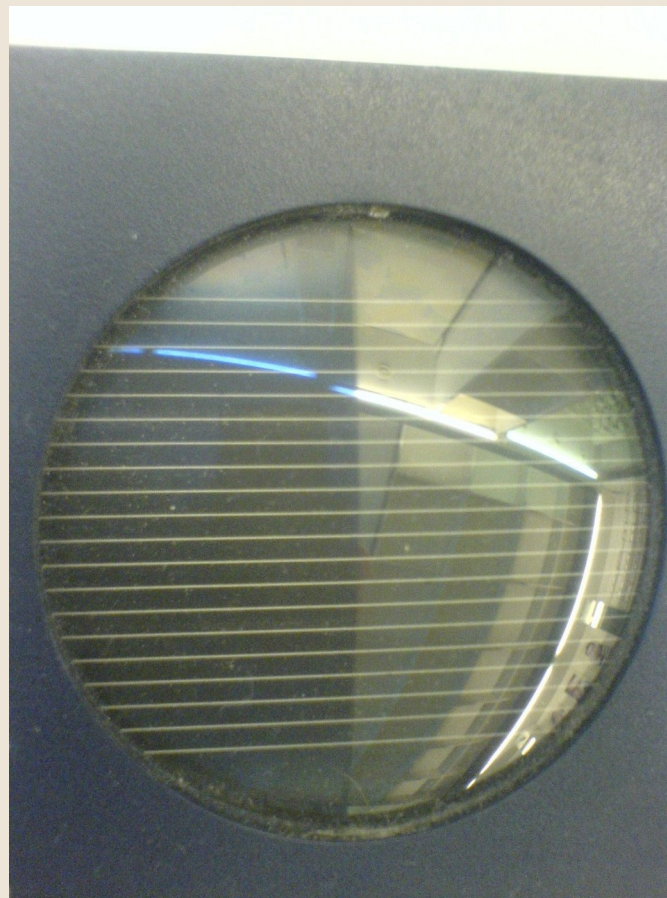
$$d = 560 \times 10^{-9}/4 \times 1.38$$

$$\underline{d = 1.01 \times 10^{-7} \text{ m}}$$

## Where you meet it — coated lenses



Uncoated and coated

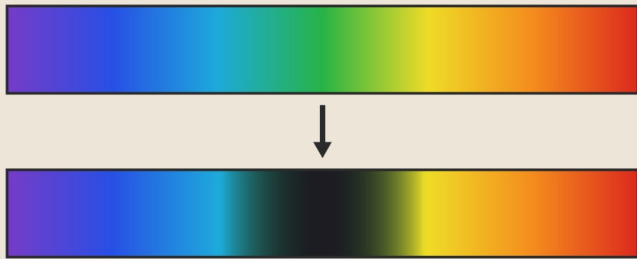


Half coated, half not

## Example answer — a purple tint

A lens coating is non-reflecting for green light. In white light the lens looks purple. Explain why. (2)

green: cancels  
red, blue: reflect



Green light is not reflected. (1)

Reflected red + blue mix: purple. (1)

Purple as one wavelength: 0 marks.

## Example answer — a blue-green tint

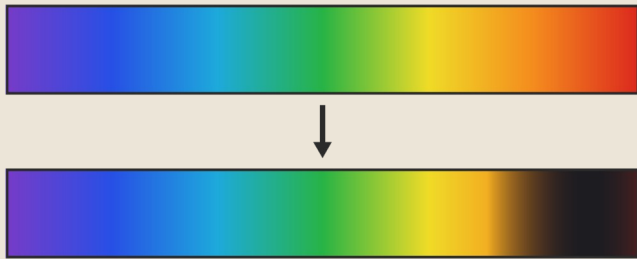
A coating is non-reflecting for red light. In white light it looks blue-green. Explain why. (2)

red: cancels

blue, green: reflect

Red is transmitted, not reflected. (1)

Blue and green are reflected. (1)



# Look back — by the end of this part...

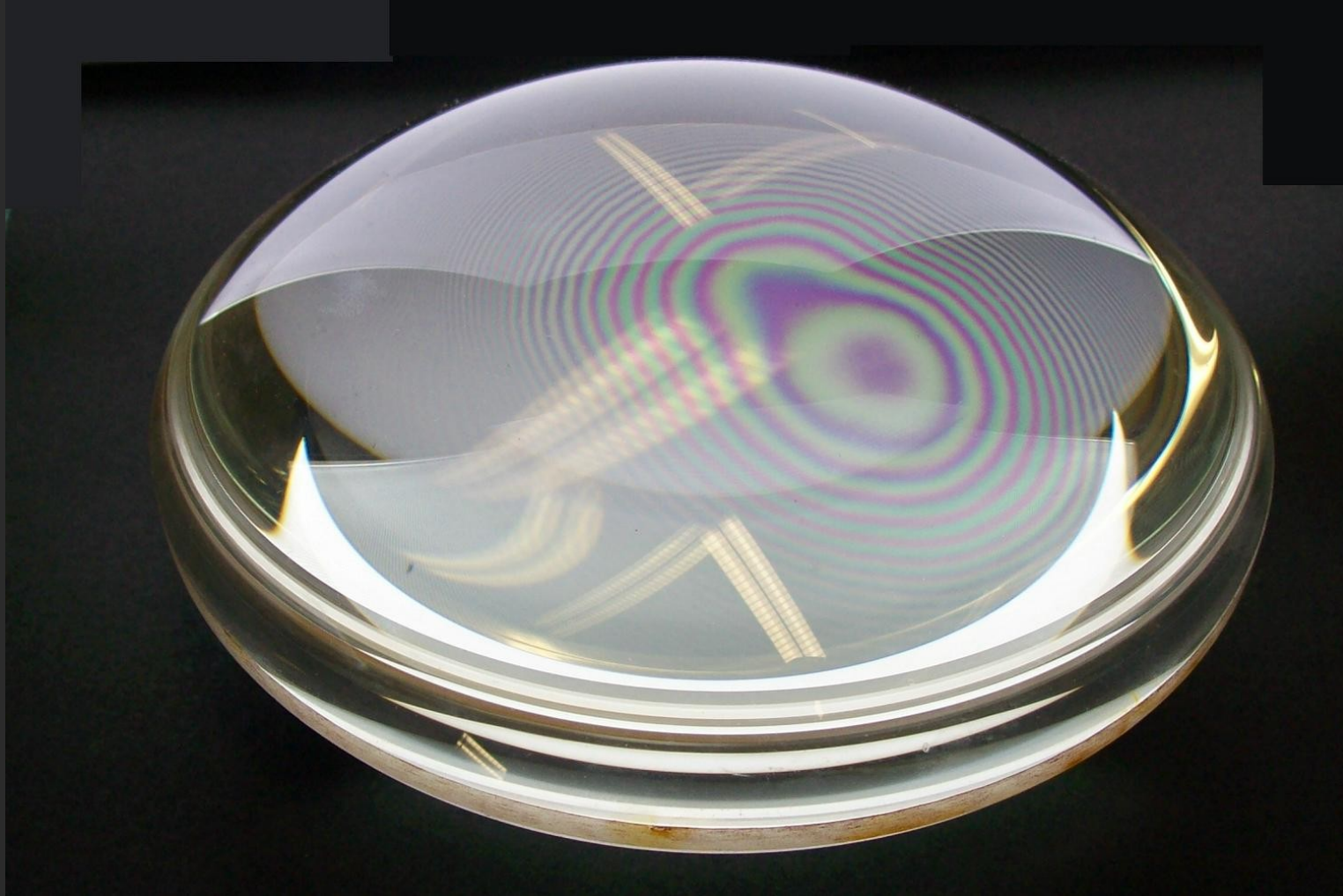


- 1 I can explain interference by division of amplitude.
- 2 I can use  $\text{opd} = m\lambda$  or  $(m + \frac{1}{2})\lambda$  to solve problems.
- 3 I can derive  $d = \lambda/4n$  for a coated lens.

# Daily Review



- 1 State what is meant by division of amplitude.
- 2 Light in glass reflects off air. State the phase change.
- 3 State the opd for destructive interference.
- 4 Why does a coated lens work for one wavelength only?
- 5 State the relationship for a coated lens thickness.



Two lenses touching, with a thin gap of air. Where do the rings come from?

### PART 3

# Can light measure a wire thinner than a hair?

Trap a wedge of air between two glass plates.

## By the end of this part...

- 1 I can describe how wedge fringes are made.
- 2 I can use  $\Delta x = \lambda l / 2d$  to solve problems.
- 3 I can state and justify how  $\Delta x$  changes.

# Wedge fringes

## Air wedge

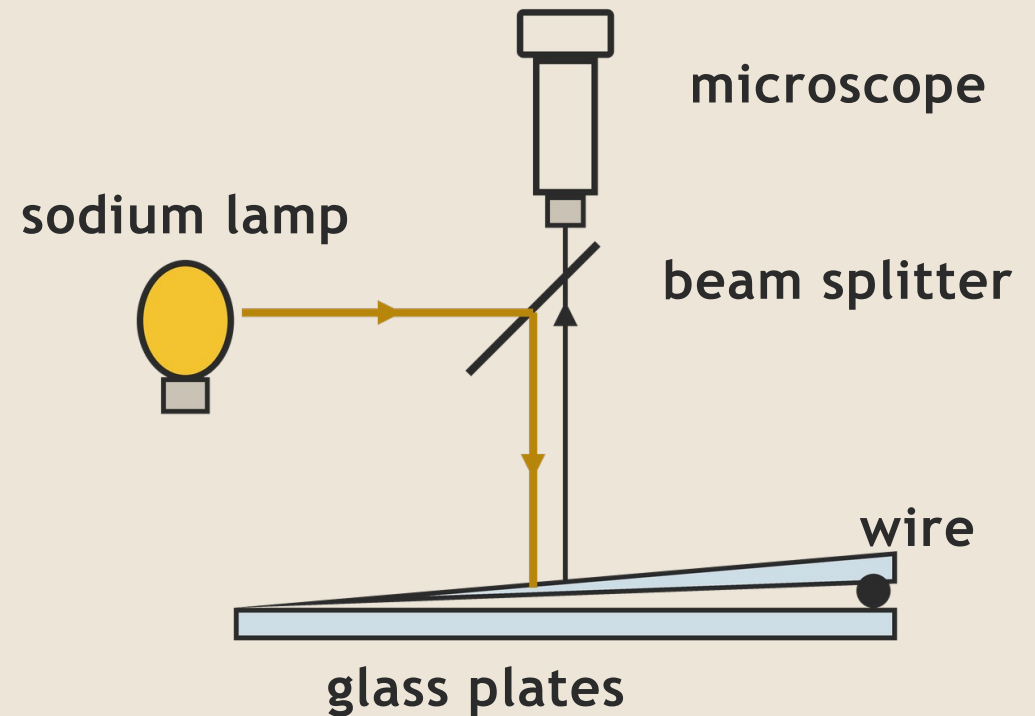
Plates touch at one end only.

## Two rays

Top and bottom of the air gap.

## Bottom reflection

$\pi$ : dark where plates touch.

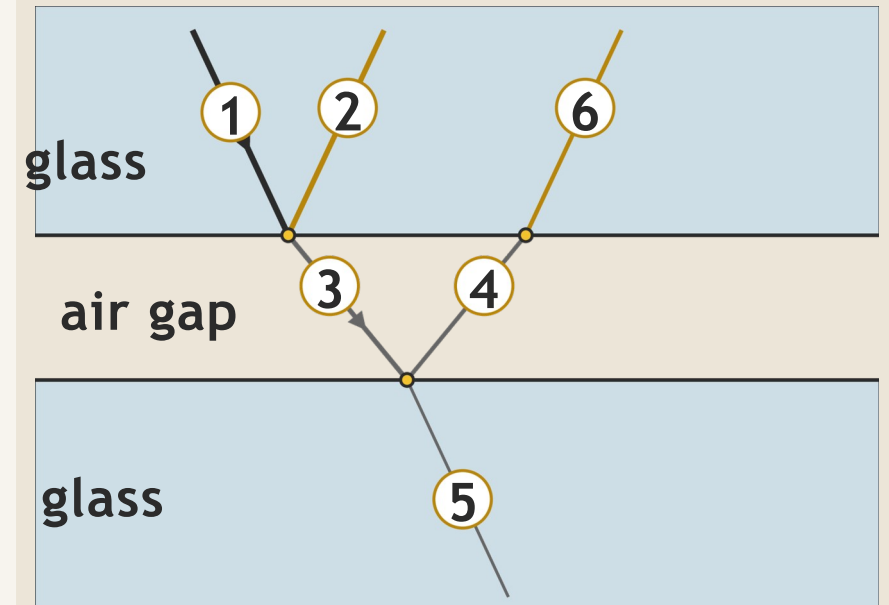


# Inside the air wedge



Dark where plates touch: ray 4 has  $\pi$ .

| ray                         | phase change |
|-----------------------------|--------------|
| 1 incident                  | —            |
| 2 reflected off less dense  | 0            |
| 3 refracted into less dense | 0            |
| 4 reflected off more dense  | $\pi$        |
| 5 refracted into more dense | 0            |
| 6 refracted into more dense | 0            |



# Wedge fringe separation



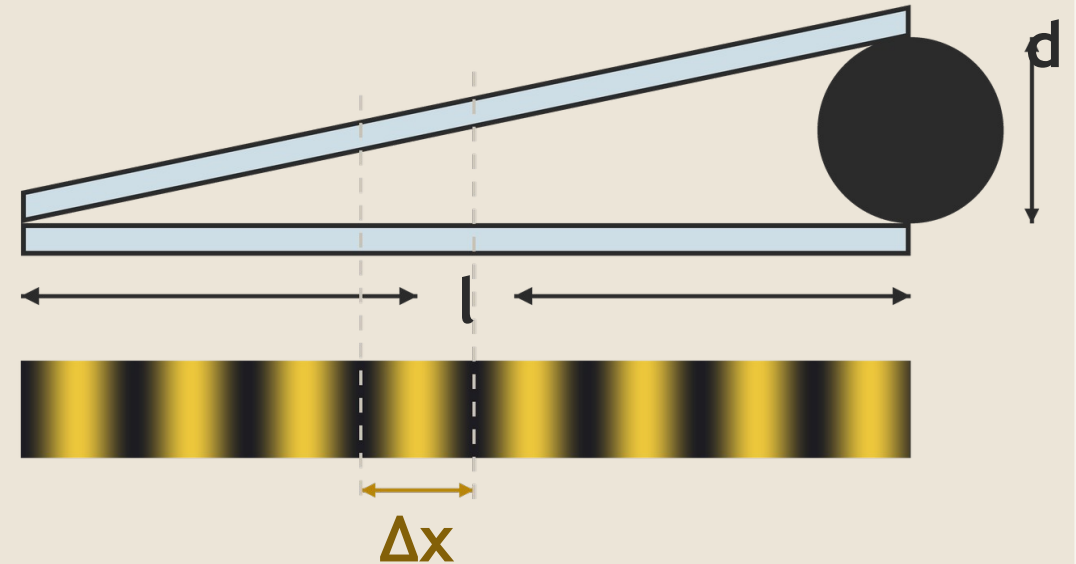
$$\Delta x = \lambda l / 2d$$

$\Delta x$ : fringe separation (m)

$\lambda$ : wavelength (m)

$l$ : length of the wedge (m)

$d$ : thickness at the wire (m)



For interest — where  $\Delta x = \lambda l / 2d$  comes from

**One fringe on**

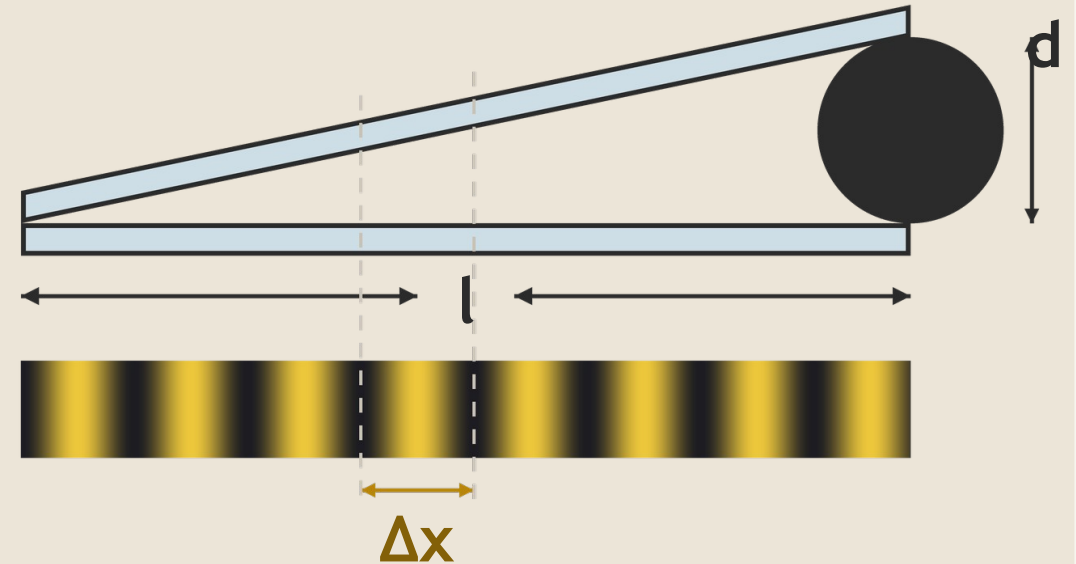
The air gap grows by  $\lambda/2$ .

**Same slope**

$$(\lambda/2) \div \Delta x = d \div l$$

**Rearrange**

$$\Delta x = \lambda l / 2d$$

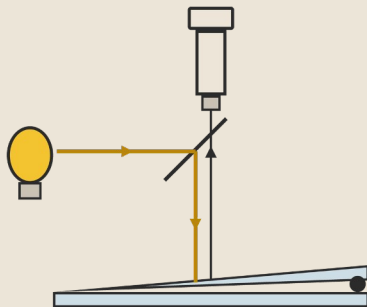


## Example answer – which division?

Sodium light reflects from an air wedge. Fringes are seen. State whether they are made by division of amplitude or by division of wavefront. (1)

two surfaces  
one ray split

Division of amplitude. (1)



## Worked example 4 – wire diameter

Plates 60.0 mm long touch at one end; a wire holds the other apart. In 589 nm light, 15 fringe separations measure 1.80 mm. Determine the wire diameter. (4)

$$15\Delta x = 1.80 \times 10^{-3} \text{ m}$$

$$l = 60.0 \times 10^{-3} \text{ m}$$

$$\lambda = 589 \times 10^{-9} \text{ m}$$

$$d = ?$$

$$\Delta x = 1.80 \times 10^{-3} \div 15 = 1.20 \times 10^{-4} \text{ m}$$

$$\Delta x = \lambda l / 2d$$

$$1.20 \times 10^{-4} = 589 \times 10^{-9} \times 60.0 \times 10^{-3} / 2d$$

$$\underline{d = 1.47 \times 10^{-4} \text{ m}}$$

## Example answer – change $d$

A current heats the wire and it expands.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

$d$  increases

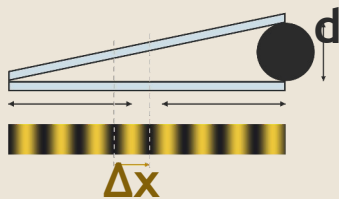
$\lambda$  unchanged

$l$  unchanged

Fringe separation decreases. (1)

$d$  increases;  $\lambda$ ,  $l$  unchanged. (1)

No reason given: 0 marks.



## Example answer – change $\lambda$

Blue light replaces the sodium light.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

blue: shorter  $\lambda$

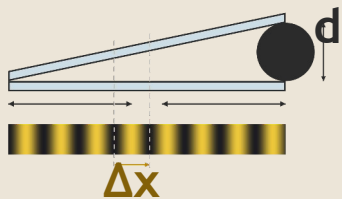
$l$  unchanged

$d$  unchanged

Fringe separation decreases. (1)

$\lambda$  decreases;  $l$ ,  $d$  unchanged. (1)

No reason given: 0 marks.



## Example answer – change $l$

The wire is moved closer to where the plates touch.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

$l$  decreases

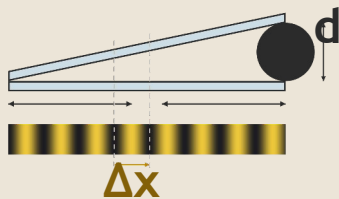
$\lambda$  unchanged

$d$  unchanged

Fringe separation decreases. (1)

$l$  decreases;  $\lambda$ ,  $d$  unchanged. (1)

No reason given: 0 marks.



# For interest – Newton's rings

## Each ring

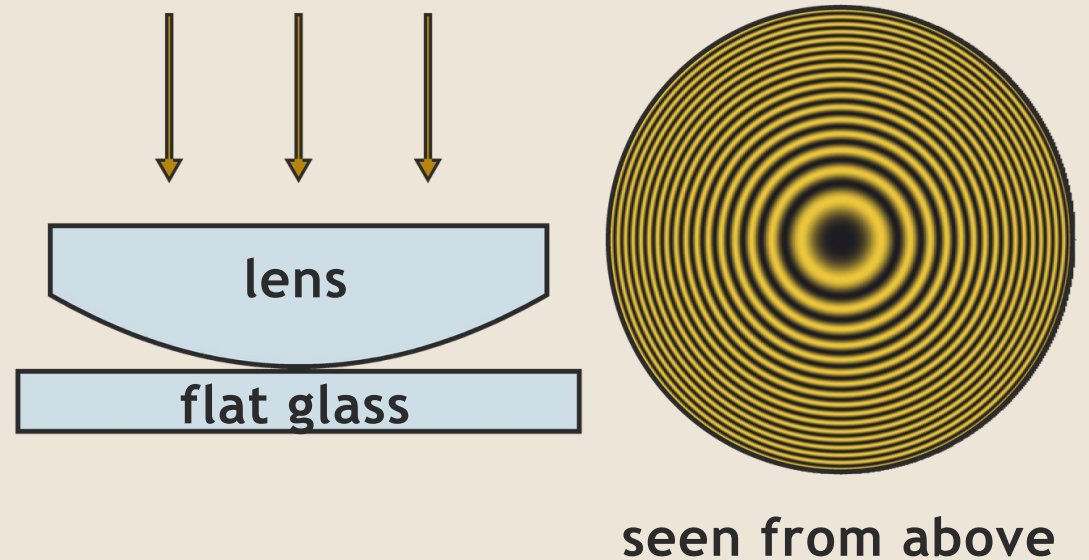
One thickness of air.

## Dark centre

Touching: phase change =  $\pi$ .

## Newton

Measured; Young explained.



# About Isaac Newton

## Who

Mathematician, physicist.

## Born

1643, Lincolnshire.

## Died

1727, London.

## Later life

Ran the Royal Mint.



# Isaac Newton – discoveries and theories

## Motion

Three laws of motion.

## Gravity

Universal gravitation.

## Light

White light: all colours.

## Mistake

Light as particles.



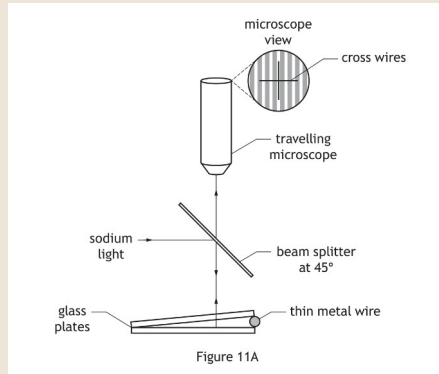
His prism experiment, rebuilt.

# Look back — by the end of this part...

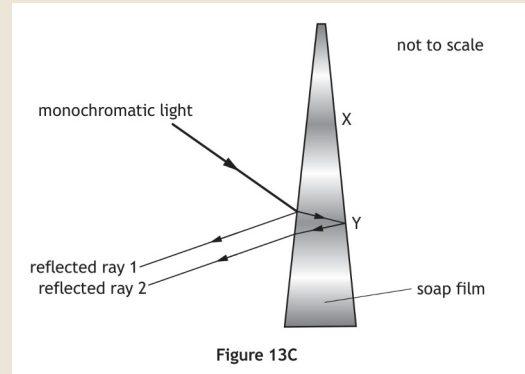


- 1 I can describe how wedge fringes are made.
- 2 I can use  $\Delta x = \lambda l / 2d$  to solve problems.
- 3 I can state and justify how  $\Delta x$  changes.

# Division of amplitude in the exam



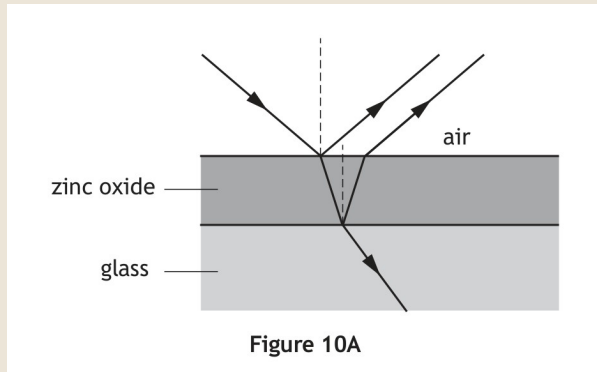
2018 Q11



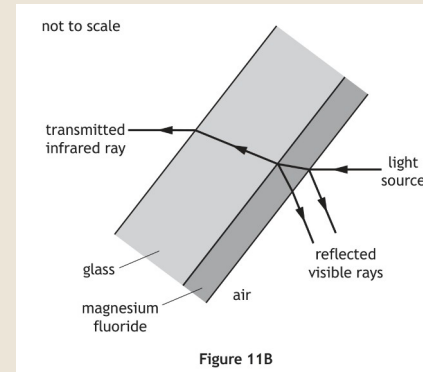
2019 Q13(c)



Past paper  
finder



2020 Q10



2023 Q11

2025 Q11(b)-(c)  
is for your test.

# Reading and homework



## Scholar Topic 7 test

Scholar – Topic 7 Division of amplitude  
Scholar – Topic 7 end-of-topic test  
BrightRed Study Guide pp. 82-83  
Past papers – 2018, 2019, 2020, 2023



Scholar

## Errors that cost marks

- Give a phase change as  $\pi$  rad, or 0.
- gpd in a film is 2d: down and back up.
- Divide by the number of fringe separations, not fringes.
- "Decreases" alone scores 0 — justify it.
- Purple is a mix of colours, never one wavelength.

## Check yourself

- 1 Calculate an opd from  $n$  and a film thickness.
- 2 State the phase change at each boundary of a coated lens.
- 3 Explain interference by division of amplitude.
- 4 Derive  $d = \lambda/4n$ , one line at a time.
- 5 Find a wire diameter from  $N$  wedge fringe separations.

## Experiments you could try

- Soap film on a wire loop: watch it drain and go black at the top.
- Two microscope slides and a hair: count the fringes in sodium light.
- Tilt coated glasses or a camera lens: look at the reflection colour.
- A drop of oil on water in a black tray: watch the colours move.
- Photograph your fringes with a ruler in shot and measure  $\Delta x$ .

## Course specification — 1 of 3

- Use of an appropriate relationship to solve problems involving optical path difference opd, geometrical path difference gpd and refractive index.  $\text{opd} = n \times \text{gpd}$
- Knowledge that a wave experiences a phase change of  $\pi$  when it is travelling in a less dense medium and reflects from an interface with a more dense medium.
- Knowledge that a wave does not experience a phase change when it is travelling in a more dense medium and reflects from an interface with a less dense medium.

## Course specification – 2 of 3

- Explanation of interference by division of amplitude, including optical path length, geometrical path length, phase difference, and optical path difference.
- Knowledge of thin-film interference and wedge fringes.
- For light interfering by division of amplitude, use of an appropriate relationship to solve problems involving the optical path difference between waves, wavelength and order number.  
 $\text{opd} = m\lambda$  or  $(m + \frac{1}{2})\lambda$  where  $m = 0, 1, 2...$

## Course specification — 3 of 3

- Knowledge that a coated (bloomed) lens can be made non-reflective for a specific wavelength of light.
- Derivation of the relationship  $d = \lambda/4n$  for glass lenses with a coating such as magnesium fluoride.
- Use of appropriate relationships to solve problems involving interference of waves by division of amplitude.  $\Delta x = \lambda l/2d$   
 $d = \lambda/4n$

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